Philosophical Foundations — Module 01: Systems — Study Questions

1. How does Active Inference relate to Descartes' mind-body problem? Why is the Markov Blanket a *dissolution* of the problem rather than a solution?
2. How does the generative model correspond to Kant's transcendental categories? What is the Active Inference equivalent of Kant's claim that we never access "the thing-in-itself"?
3. What is the difference between classical Bayesian epistemology and Active Inference's variational inference? Why is the shift from "ideal" to "approximate" inference philosophically significant?
4. Explain process epistemology. How does the FEP replace static notions of knowledge/justification/truth with dynamic ones (model-evidence alignment, predictive accuracy, precision)?
5. What is the circular epistemology problem? The model defines what counts as evidence, and evidence updates the model. How does the FEP embrace this circularity?
6. How does the FEP's circularity relate to (a) Quine's naturalized epistemology, (b) Heidegger's hermeneutic circle, and (c) Maturana & Varela's autopoiesis? In what ways is each comparison illuminating, and where does each break down?
7. Is the Markov Blanket a *discovery* (realism) or a *construction* (anti-realism)? What would a structural realist say? Argue each position with examples.
8. What is "self-evidencing" (Hohwy)? How does it provide a formal criterion for the boundary between "mere self-organization" and genuine cognition?
9. How does the FEP relate to pragmatism? William James argued that truth is "what works." Active Inference implies that good models are those that minimize free energy. Are these the same claim?
10. What is the relationship between the complexity penalty in free energy and Occam's Razor? Does the brain automatically implement Occam's Razor? Is this normatively justified?
11. How does the FEP handle the problem of induction (Hume)? If the brain's generative model can only learn from past experience, what justifies its predictions about the future?

12. Can the FEP account for scientific revolutions (Kuhn)? Is a paradigm shift analogous to Bayesian Model Reduction or to Bayesian Model Selection? What would a Kuhnian revolution look like in Active Inference terms?
13. Does the FEP commit the "mereological fallacy" — attributing to the brain properties that properly belong to the person? Does the brain "infer" or does the person?
14. How does the FEP relate to enactivism (Varela, Thompson, Rosch)? Are they complementary frameworks or competitors? Where do they agree and where do they conflict?
15. Connect the FEP to Nishida's concept of "pure experience" (Japanese philosophy) or Zhuangzi's "forgetting of self" (Chinese philosophy). Does Active Inference have non-Western philosophical roots? Should it?
16. How does the concept of "model evidence" relate to Popper's falsifiability criterion? Can a model be falsified in Active Inference, or only have lower/higher evidence?
17. If all organisms minimize free energy, does this make the FEP unfalsifiable? What would count as evidence *against* the FEP?
18. How does the FEP handle the hard problem of consciousness (Chalmers)? Does minimizing free energy explain why there is "something it is like" to be an organism, or does it explain only the functional correlates?
19. Is the FEP a theory of everything, or is it a framework within which theories can be formulated? What is the epistemological status of the FEP itself?
20. Critically evaluate: Does the FEP replace traditional epistemology or merely supplement it? What important epistemological questions does it leave unanswered?